

Project Proposal

You will need to submit a pdf document describing what you want to do. The document should contain the following sections:

- Title with project name
- Project summary/description (up to 100 words)
- Data sources table. Fill in the following table:

Data source #	Name / short description	Source URL	Type: - API - Web page - file	List of fields	Format: - json - xml - csv - sql - other	Have tried to access/ collect data with python ? yes/no	Estimated data size, number of data points you plan to use
1	U.S. Chronic Disease Indicator/ Risk factors to chronic diseases	https://healthdata.gov/CDC/US-Chronic-Disease-Indicators/dhcp-wb3k/about_data	File	Healthcare	CSV	Not yet (n)	300
2	BRFSS '23	https://www.kaggle.com/datasets/isuruprabath/brfss-2023-csv-dataset?select=BRF	File	Healthcare	CSV	Not yet (n)	300

		SS2023.csv					
3	Behavioral Risks factors/ CDC Behavioral risk factors	https://data.cdc.gov/api/v3/visualizations/hn4x-zwk7/query.json	File	Healthcare	JSON	Not yet (n)	300
4	United Healthcare / America's Health Rankings	https://api.america'shealthrankings.org/graphql	API	Healthcare	JSON	Not yet (n)	300

- Identify 3 data sets to support your project
- At least 1 data set must be collected via API or via web scraping
- What type of analysis or model you plan to do on the data (free form but ideally less than 150 words).

Title: AI for Health: Predicting Chronic Disease Risks through Machine Learning

As a public health undergrad major, I am deeply passionate about the healthcare field and now seek to leverage my foundational knowledge by applying this knowledge into the field of data science. This project aims to apply healthcare data science methods to explore behavioral factors that influence the development of chronic diseases and their impact on overall health rankings across the United States. I plan to develop a predictive model that predicts the likelihood of developing a chronic disease based on behavioral risk factors and examines how these outcomes relate to health rankings nationwide. To achieve this, I will use logistic regression as a baseline method to interpret the extent and significance of each risk factor. Ultimately this project will demonstrate my ability to leverage data-driven analysis to generate meaningful insights in improving health outcomes and addressing chronic disease risks at the population level.